

2024



Hokkaido Cultural Broadcasting Streamlines Production of News Scripts and Videos Using Amazon Bedrock to Deliver More News at a Lower Cost

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2 minutes

to generate news articles

100%

more



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Dramatically reduced production costs



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Overview

[Hokkaido Cultural Broadcasting](#) airs a variety of news and shows that showcase the nature, food, and tourist attractions of Hokkaido, Japan. Expanding its online streaming services, the station used [generative artificial intelligence \(AI\)](#) and [text-to-speech](#) from Amazon Web Services (AWS) to streamline production of news scripts and videos, resulting in a low-cost solution to deliver more news. The broadcaster also made news sources easier to search using [Amazon DynamoDB](#)—a serverless, NoSQL database service.



Opportunity | Reducing Production Costs of News Video Streaming

Established in 1972, Hokkaido Cultural Broadcasting serves its programs to the entire Hokkaido area as an affiliate of the [Fuji Television Network](#). The company strives for technological innovation while promoting the charms of Hokkaido through community-based content. The station's development initiatives on broadcasting and information processing are led by Ayuki Sugimoto in the Programming Department and Technical Lead concurrently at Hokkaido Cultural Broadcasting. "While I'm involved in data broadcasting and conventional programs, I also work on system development," says Sugimoto. "Our company doesn't have a dedicated system development department. Volunteers from the news reporting team do it."

Previously, ratings, program content, and social media information were managed in separate formats, making data aggregation and analysis difficult. As a solution, Sugimoto created a business intelligence (BI) tool using [Amazon QuickSight](#)—which powers data-driven organizations with unified BI at hyperscale—to visualize and analyze the data at a low cost. "We didn't know anything about AWS or the cloud to begin with," says Sugimoto.



“With support from AWS, we created a BI tool in just a couple of weeks. The support and plentiful hands-on materials from AWS made it easier for the beginner to get started.”

Sugimoto’s transformative digital data visualization tool was met with great acclaim and won the Japan Commercial Broadcasters Association’s Technology Award in 2023.

In the spring of 2023, Sugimoto started a new project involving news video production by the news department. “The news department wanted to publish more news online, but allocating production personnel was too expensive,” says Sugimoto. “So, we looked at ways to automate the process of creating news articles and videos.”

The news department writes articles and publishes them online by covering stories on press releases faxed by regional governments and companies daily. However, the team has a limited number of reporters, and the quality of their scripts and the time needed to create them depends on the skills of those reporters. Several more challenges prevented quick news distribution, including arranging announcers when creating audio and video based on the articles.

Solution | Streamlining News Video Production with Generative AI and Text-to-Speech Tools

Sugimoto turned to AWS to streamline the production of news articles. Hokkaido Cultural Broadcasting had converted press releases to PDF format on a multifunction printer and emailed the PDF files to the news team. Sugimoto built a solution to perform optical character recognition on these PDF files and store the text data on Amazon DynamoDB, letting the team view the releases in text on a management screen. Members of the news team cover the story based on the releases and enter additional information through the management screen so that [Amazon Bedrock](#)—which is a fully managed service that offers a choice of high-performing foundation models from leading AI companies—can use all the data to output a news script. “We’ve configured the tool to output three model scripts for each news script,” says Sugimoto. “Our reporters choose the best script and edit it to complete the article. We use Anthropic’s Claude 2.1 as the foundation model, which delivers excellent accuracy and cost performance.”

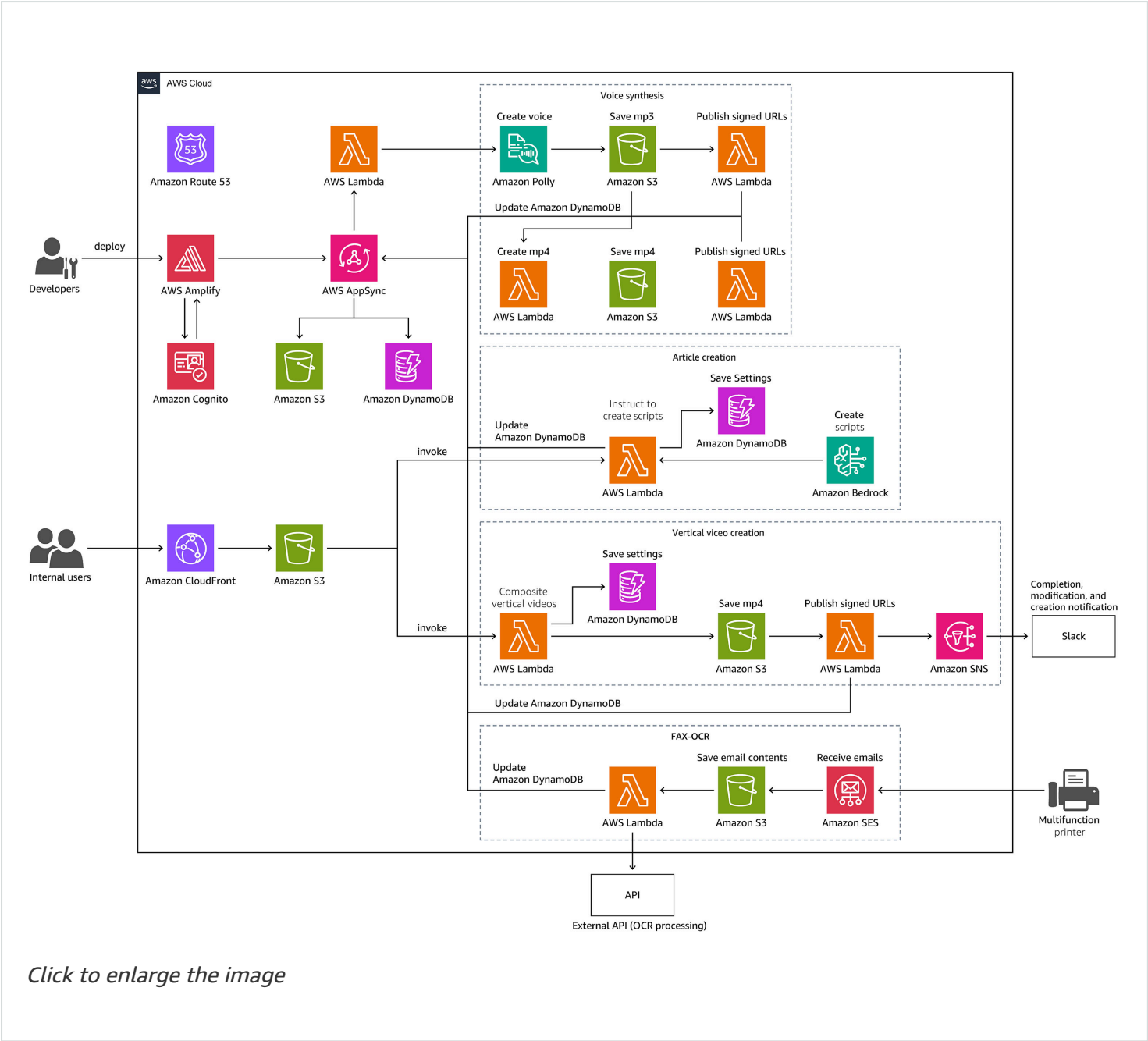
With the station receiving over 1,000 press releases every month, the new solution simplifies searches the releases. Team members can use keywords to search text information stored on Amazon DynamoDB with less effort.



Supplementing online news articles with videos adds value and opens other distribution channels, such as YouTube. Hokkaido Cultural Broadcasting therefore adopted [Amazon Polly](#), which uses deep learning technologies to synthesize natural-sounding human speech, to narrate news videos. “When synthesizing speech, you must correct misreading and intonation,” says Sugimoto. “Because it was difficult for the operations team to add SSML tags to fine-tune the narrations, we simplified things for easier control. The team can perform operations like adjusting the time between sentences by adding line breaks from the management screen.”

The team also developed a tool to more easily create vertical videos, such as YouTube Shorts. The team members would previously perform this with a video editing application, but the time needed for the task could vary depending on each member’s proficiency. The station therefore built a user-friendly, browser-based solution that lets users choose a format, generate text and videos, and share them. This mechanism is designed using an AWS serverless environment and has passed internal security regulations.

Architecture



Using AWS to help create articles has had a large effect at Hokkaido Cultural Broadcasting. “Amazon Bedrock helps us to automatically generate an article in about 2 minutes for only a few extra yen per article,” says Sugimoto. “The quality of the articles is that of journalists with a couple of years’ experience, and the news department can write articles significantly faster now. We can produce a short video for a single article in minutes, which is less than 20 percent of the old time of 30 minutes to an hour. And we’re distributing our news content through more channels by using videos with synthesized speech. We’re able to provide 100–120 additional pieces of content per month without a significant cost, resulting in a higher value.”

In October 2023, Hokkaido Cultural Broadcasting won the gold prize at the 31st Fuji Network System Technical Fair among the Fuji Television affiliates for initiatives to simplify the creation of news assets and to automate the organization of inbound faxes. Following these honors, the company has been contacted by several other stations with inquiries and requests for visits. “The board of directors decided to set up a team to use generative AI for improving operations,” says Takashi Nakagawa, director and head of the Content Division at Hokkaido Cultural Broadcasting. “This decision was led by both internal and external recognition of Sugimoto’s untiring work. There have been projects that are difficult to test at flagship stations but easily implementable at local broadcasters, so I want these types of initiatives to continue.” Sugimoto will continue to be the key figure in Hokkaido Cultural Broadcasting’s plans for building business-enhancing systems.

“There are very few hurdles to experimenting on AWS, and the low initial cost is incredibly helpful, especially for small- and medium-sized businesses in provincial areas,” says Nakagawa. He is particularly interested in the active exchange of information at AWS user groups, such as JAWS-UG Sapporo and JAWS-UG Media. “These groups hold meetups where people from different organizations gather. There weren’t many places where we could network with external users and help each other previously, so I hope these kinds of activities expand even further.”

About Hokkaido Cultural Broadcasting

Established in 1972, Hokkaido Cultural Broadcasting serves its programs to the entire Hokkaido area. The station provides Fuji News Network and Fuji Network System programming and also offers a variety of local news and the nature, food, and tourist attractions of Japan’s northern island. In recent years, the broadcaster has made efforts in online streaming on platforms such as YouTube and distributes a local perspective of Hokkaido to a broader audience.



Takashi Nakagawa



Ayuki Sugimoto

AWS Services Used

AWS Lambda

AWS Lambda is a compute service that runs your code in response to events and automatically manages the underlying compute resources, making it the fastest way to turn an idea into a modern, production, serverless application.



applications.

[Learn more »](#)

Amazon Bedrock

Amazon Bedrock is a fully managed service that offers a choice of high-performing foundation models (FMs) from leading AI companies like AI21 Labs, Anthropic, Cohere, Meta, Mistral AI, Stability AI, and Amazon through a single API, along with a broad set of capabilities you need to build generative AI applications with security, privacy, and responsible AI.

[Learn more »](#)

Amazon Polly

Amazon Polly uses deep learning technologies to synthesize natural-sounding human speech, so you can convert articles to speech.

[Learn more »](#)

Amazon DynamoDB

Amazon DynamoDB is a serverless, NoSQL database service that enables you to develop modern applications at any scale.

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